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Final Paper

Introduction:

We know that demographics play a significant role in shaping political representation and legislative effectiveness in the U.S. House of Representatives. However, we don't know exactly how characteristics such as race, gender, age, and political stances influence the ability of House members to sponsor bills and affect the legislative process. So, I conducted research to explore the relationships between demographic features and the legislative effectiveness of House members, with a specific focus on race and bill sponsorship. This research aims to shed light on how these factors may intersect with the legislative process, highlighting potential disparities in representation and the ability to sponsor legislation across different racial groups.

As demographic diversity in Congress has grown, it is increasingly important to examine whether different groups are equally able to influence legislative processes. Research in this area can uncover systemic biases that affect the ability of members from marginalized racial groups to sponsor bills. The social significance of this question is profound, as it touches on equity, representation, and the power dynamics that govern legislative success. By analyzing the relationship between race and bill sponsorship, we can evaluate whether certain racial groups face barriers in shaping legislative agendas or if their bills are disproportionately overlooked.

For this study, data was collected on all 437 members of the U.S. House of Representatives, including demographic variables such as race, gender, party affiliation, and other political and personal characteristics. The specific focus of my analysis is on the relationship between racial identity and the number of bills sponsored. The research question asks whether racial identity influences the number of bills a House member sponsors. The null hypothesis states that there is no significant difference in the mean number of bills sponsored among different racial groups, while the alternative hypothesis posits that at least one racial group's mean differs significantly from the others. To test this hypothesis, I used a one-way ANOVA test, which is appropriate when comparing the means of a quantitative variable (number of bills sponsored) across multiple categorical groups (racial identities). The ANOVA test was ideal because it allows me to determine whether any statistically significant differences exist in the number of bills sponsored among the seven racial groups.

The results of my analysis revealed a statistically significant relationship between racial identity and the number of bills sponsored by members of the U.S. House of Representatives.

These findings are important because they highlight potential disparities in legislative behavior that may be influenced by systemic or structural factors. Understanding whether racial identity correlates with bill sponsorship can help us evaluate broader patterns of representation, legislative opportunities, and systemic challenges that lawmakers from marginalized groups may face.

Data and Methods

The first step in a statistical study is to collect a sufficient sample size. To avoid inadvertently overrepresenting or underrepresenting certain demographic groups within the House, we decided to use a census approach, which involves collecting data from every member of the "population" —in our case, the entire House of Representatives. In our study, the unit of analysis is each individual House member, and as mentioned, the population encompasses all current representatives in the House, providing us with a sample size of 437. Given the legal limit of 435 members in the House of Representatives, we figured that there was a slight error in our sample collection. Upon further investigation, we found that one of the extra subjects was a deceased former member, and the other was Jennifer Gonzalez, Puerto Rico's U.S. Representative. However, Gonzalez is not a voting member of the House. Data for these variables were collected from publicly available congressional records, ensuring an accurate and standardized measurement of legislative activity.

Descriptive Statistics for Variables Used in the Analysis

	Mean	SD	Min	Max	Description
bills	85.76	127.23	0	1109	amount of bills they have sponsored
race					racial/cultural identification
Black/African American	0.141				Representative is Black/African American
Native American/Alaska Native	0.007				Representative is Native American/Alaska Native
Asian/Native Hawaiian	0.035				Representative is Asian/Native Hawaiian
Pacific Islander	0.00234				Representative is Pacific Islander
Bi-racial/Multi-racial	0.00469				Representative is Bi-racial/Multi-racial
White	0.718				Representative is White
Hispanic-Latino	0.0915				Representative is Hispanic-Latino
n = 426					

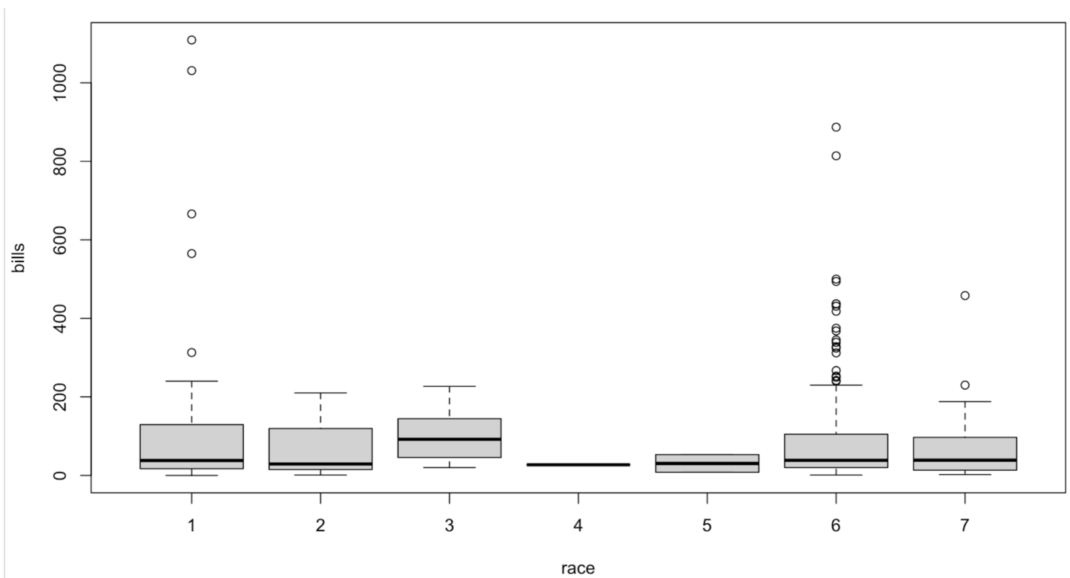
In my testing, I used a complete case analysis, which means I used observations with no missing data for the variables included. Out of the 437 total observations in the dataset, 426 observations had complete data and were included in the analysis. The remaining 11 observations had missing data on one or more variables used in the model and were excluded.

We operationalized the concept of race by treating it as a categorical variable with seven distinct categories. Each category corresponds to a racial or ethnic identity, which has been assigned a numerical code for analysis purposes: 1= Black/African American, 2= Native American/Alaska Native, 3 = Asian/Native Hawaiian, 4 = Pacific Islander, 5 = Bi-racial/Multi-racial, 6 = White, and 7 = Hispanic/Latino. By assigning these numerical codes, race can be quantitatively analyzed in statistical tests such as analysis of variance, or ANOVA. Additionally, we ensured that each House member is placed in only one category, the variable becomes mutually exclusive, allowing for clear group comparisons in our testing.

The variable "bills," is a discrete quantitative variable as it represents a count of the number of bills a representative has sponsored during their term. Discrete variables are those that can take on a finite number of values and are often counts or integers. In this case, the number of bills was expressed as a whole number (e.g., 458, 77, 19, etc.). The total number of bills that each representative sponsored was recorded, which helped us to measure the degree of legislative engagement and activity.

The analytic strategy for this research involves using a one-way ANOVA (Analysis of Variance) to determine if there are any statistically significant differences in the mean number of bills sponsored for each racial group. By comparing the mean number of bills sponsored for each racial group, I am to explore whether racial identity influences the number of bills sponsored by members of the U.S. House of Representatives. The null hypothesis (Ho) is that there is no significant difference in the mean number of bills sponsored across different racial groups, and my research hypothesis (Ha), suggests that at least one racial group differs from others in terms of the number of bills they sponsor, indicating there to be a significant correlation between racial identity and legislative engagement.

<u>Results:</u>



The boxplot of bills sponsored by race reveals that while the median number of bills, there is substantial variability within groups, particularly among Black/African American (Group 1) and White (Group 6) representatives, where notable outliers are present. These outliers, including individuals sponsoring over 600 or 1000 bills, may disproportionately influence the results, contributing to inflated between-group variance. Groups with smaller sample sizes, such as Asian/Native Hawaiian (Group 3), Pacific Islander (Group 4), and Bi-racial/Multi-racial (Group 5), exhibit minimal spread, which could reflect sample size limitations rather than true variability. Overall, the high within-group variance compared to relatively small differences in group medians suggests that while race plays a role, other factors likely influence the number of bills sponsored.

ANOVA Test:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
bills	1	17.5	17.530	5.09	0.0246 *
Residuals	434	1494.7	3.444		

The one-way ANOVA test reveals a statistically significant relationship between race and the number of bills sponsored by members of the U.S. House of Representatives ($F = 5.09$, $p = 0.0246$) at the 5% significance level. This allows us to reject the null hypothesis, indicating differences in the mean number of bills sponsored across racial groups. However, the low R-squared value of 1.15% suggests that race explains only a small portion of the variation in bill sponsorship, highlighting that while the relationship is statistically significant, its practical impact is minimal.

Discussion

The effect observed in the one-way ANOVA analysis is statistically significant ($p = 0.0246$), indicating that race is associated with differences in the number of bills sponsored by members of the U.S. House of Representatives. However, this relationship is likely correlational rather than causal, as race itself does not directly determine legislative behavior. A potential confounding variable—a variable that influences both the independent (race) and dependent (bills sponsored) variables—could be seniority or party affiliation. Seniority may explain variation in legislative output, as more experienced lawmakers may sponsor more bills, while party affiliation could reflect differences in political priorities and institutional support for bill sponsorship. These factors likely interact with racial identity in complex ways, influencing legislative behavior and outcomes.

The real-world implications of these findings are significant despite the small effect size. The statistical significance of the relationship suggests that race, among other factors, may play a role in shaping legislative activity. This raises concerns about systemic barriers that lawmakers from marginalized racial groups may face, such as unequal access to political networks, resources, or leadership positions that support bill sponsorship. Understanding these dynamics is critical for evaluating the equity of political representation and ensuring that all members of Congress, regardless of race, have equal opportunities to influence legislative agendas. While race alone accounts for only 1.15% of the variation in bill sponsorship, it highlights the need to further investigate how structural and institutional factors impact legislative engagement.

There are several limitations to this research that must be addressed. First, the sampling strategy—though comprehensive—may still contain biases, such as underreporting of certain demographic nuances or the inclusion of representatives with disproportionately high or low sponsorship counts (outliers). The boxplot shows significant outliers in Groups 1 (Black/African American) and 6 (White), where some individuals sponsored a much higher number of bills than the group medians. Since outliers can disproportionately affect groups means, potentially leading to a false positive result, otherwise known as a Type 1 error. Second, the operationalization of race into mutually exclusive categories, while useful for statistical testing, oversimplifies the fluid and intersectional nature of racial identity. Additionally, the analytic strategy (one-way ANOVA) assumes homogeneity of variance across groups, which may be violated given the observed spread and outliers in the data. These limitations could impact the robustness of the findings and warrant caution in generalizing the results.

Future research should explore additional variables that may better explain the variance in legislative behavior. For instance, incorporating seniority, party affiliation, or district characteristics, as control variables in a multivariate analysis could provide a clearer picture of the factors influencing bill sponsorship. Additionally, calculating the difference in group means could enhance the analysis and serve as a strong foundation for further research and conclusion. This approach would help to better understand where the significant differences observed in the ANOVA test are coming from. Simply knowing that a difference exists is important, however, calculating the actual difference between group means can quantify how large or small the disparity is, which can help contextualize its real-world impact.

Overall, conducting this research has been a valuable and enlightening experience. Through this project, I have gained a deeper understanding of statistical methods, particularly the ANOVA test, and how to apply these tools to real-world questions of equity and representation. While the results showed a small effect size, they reinforced the importance of examining systematic inequalities in relation to legislative behavior. Moving forward, I am excited to continue exploring questions of political representation and how quantitative methods can uncover patterns in governance.